

SMART INDIA HACKATHON 2026

Problem statement ID – 26176

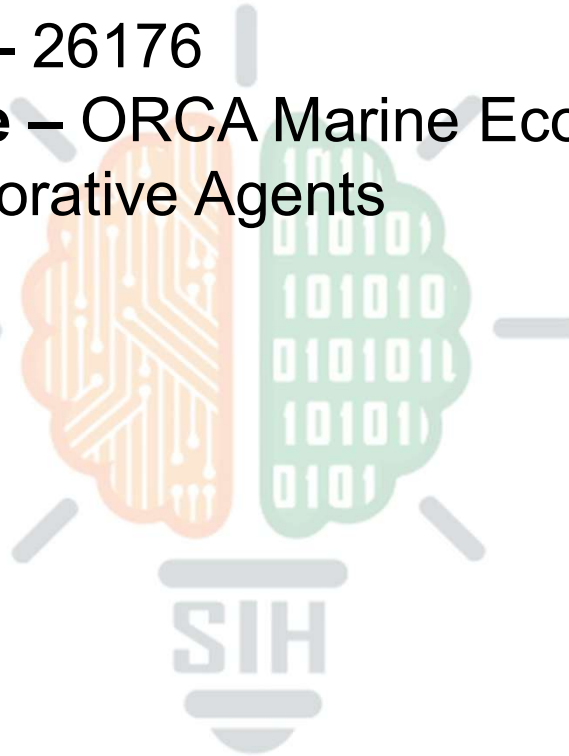
Problem statement title – ORCA Marine Ecosystem

Reasoning with Collaborative Agents

Theme – Miscellaneous

PS category – Software

Team name – Cyan



The problem: Marine information is fragmented across satellite, oceanographic, weather and geospatial sources, making it difficult for marine users to turn raw data into timely decisions.

PROPOSED SOLUTION

ORCA - AI POWERED CONVERSATIONAL MARINE INTELLIGENCE

progressive web app(PWA)

1. ORCA receives

- Voice / text query
- User's GPS location
- Time & contextual information

2. ORCA reasons across

- Weather
- Ocean conditions
- Satellite data
- Fishing data
- Geospatial boundaries

3. ORCA delivers

Visual Marine Map

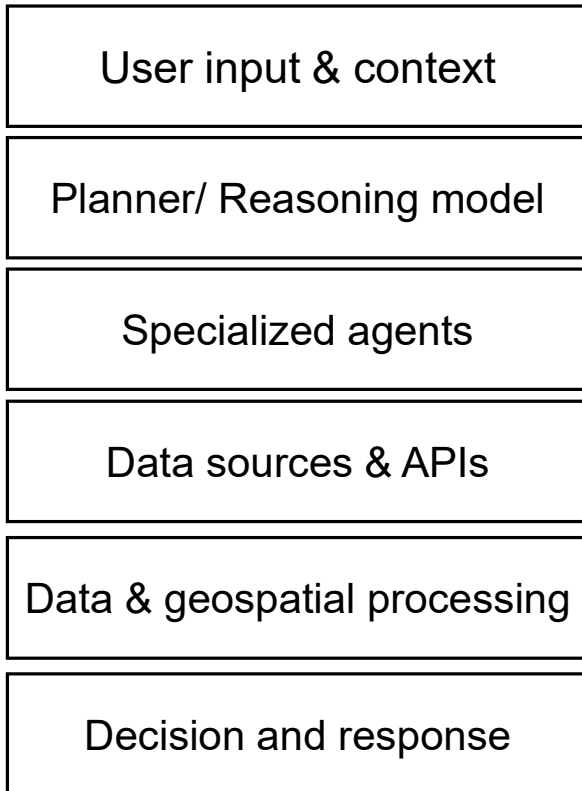
- PFZs highlighted
- Safe / unsafe areas
- Restricted & protected zones
- Recommended route / destination

Conversational Explanation

- What is happening
- What to do
- Why ORCA recommends it
- Supporting data

ORCA unifies fragmented marine data with conversational AI to deliver visual, context-aware and explainable marine decisions.

TECHNICAL APPROACH



6. Decision & Response

Combines agent results

Brings together the relevant findings from each specialized agent.

Final Reasoning

Checks the collected information against the user's location, time and request.

Actionable Recommendation

Provides a clear answer such as a fishing zone, safe/unsafe assessment or recommended route.

Visual Output

PFZ and favourable areas

Hazards and restricted zones

Recommended routes

Conversational Response

Explains **what to do and why**

Turns the collected data into an explainable decision for the user.

Tech stack:

Python, FastAPI, Open-source LLM, Open-source STT/TTS models, PostgreSQL + PostGIS, PWA (HTML/CSS/JavaScript), REST APIs, MapLibre/OSM

FEASIBILITY AND VIABILITY

What makes it feasible?

Existing Data Sources

ISRO/MOSDAC, INCOIS, IMD and public geospatial datasets already provide much of the required marine information.

Open-source Technology

Open-source LLMs, speech models and standard Python/web technologies reduce development and licensing overhead.

Modular Architecture

Each agent handles a specific task, so components can be developed and tested independently.

Offline Support

Maps and relevant marine data can be cached before going offshore, allowing the system to remain useful with poor connectivity.

Key challenges

Data availability & different formats

Marine data comes from multiple sources and may not follow the same structure.

Network limitations

Offshore users may have weak or no connectivity.

Regional-language accuracy

Marine queries need to work reliably across Indian languages.

AI reliability

Recommendations need to be based on actual marine data rather than unsupported LLM responses.

How we handle them

Standardized data layer + fallback sources

Caching + offline maps/data

Language-specific testing and prompts

Evidence-based responses with source data shown to the user

IMPACT AND BENIFITS

For fishermen

Know where to go

See nearby potential fishing zones and favourable marine conditions on the map.

Know when not to go

Get warnings about rough seas, storms, lightning and other hazards before heading out.

Plan the trip better

Use marine conditions and route information to avoid unnecessary travel and fuel use.

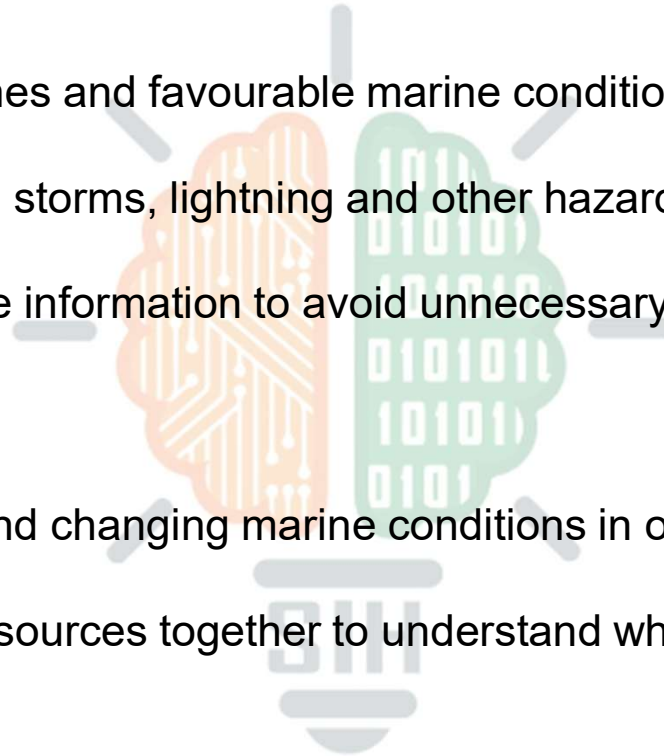
For Other Marine Users

Authorities

See hazards, restricted areas and changing marine conditions in one place.

Researchers & planners

Bring information from different sources together to understand what is happening in the ocean.



RESEARCH AND REFERENCES

ISRO / MOSDAC – mosdac.gov.in

INCOIS – incois.gov.in

IMD – mausam.imd.gov.in

Bhuvan / Marine Regions – bhuvan.nrsc.gov.in

GEBCO – gebco.net

AI4Bharat – ai4bharat.iitm.ac.in

OpenStreetMap / MapLibre – maplibre.org

LangGraph – langchain.com

